React Learning Guide

React is a JavaScript library used for building user interfaces. It allows developers to create reusable UI components and manage application state efficiently.

What is React?

React was created by Facebook. It uses a component-based architecture and virtual DOM for efficient UI rendering.

React Components

Components are reusable building blocks in React. There are functional components and class components, but modern React mostly uses functional components.

React Hooks

Hooks allow functional components to use state and lifecycle features. Common hooks include useState, useEffect, useContext, useMemo, and useRef.

useState Hook

useState allows components to store and update state values. Example: const [count, setCount] = useState(0);

useEffect Hook

useEffect is used for side effects such as API calls, timers, and event listeners. It runs after component rendering.

Props in React

Props are used to pass data from parent components to child components.

React Router

React Router enables navigation between pages in single-page applications.

State Management

Large React applications often use Redux, Zustand, or Context API for global state management.

Concept	Purpose
useState	Manage local state
useEffect	Handle side effects
Props	Pass data between components
React Router	Navigation between pages